City of Tacoma Lidar 2024 and Orthoimagery Collection Project Addendum

Land Cover and Forestry products

March 14, 2025

Prepared For:

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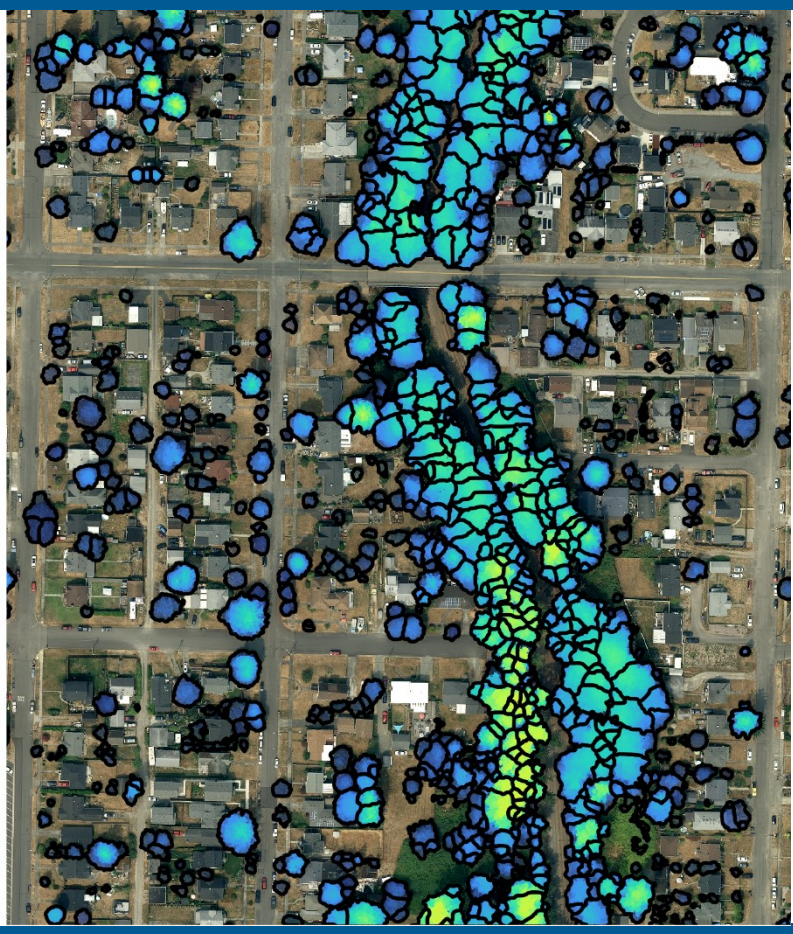

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City of Tacoma Lidar 2024 and Orthoimagery Collection Project

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1.0 INTRODUCTION

In July 2024, NV5 was contracted by Washington Department of Natural Resources (WADNR) to collect Light Detection and Ranging (lidar) data and digital imagery in the late summer of 2024 for the City of Tacoma site in Washington. Data were collected to aid WADNR in conducting hazard and natural resource mapping as well as other assets and resource analyses.

This report accompanies the delivered land cover and tree segmentation products with contract specifications, processing methods, and analysis. It complements a separate report delivered with the lidar products titled "City of Tacoma, Washington 2024 Lidar Technical Data Report" submitted December 09, 2024.

Figure 1 shows the City of Tacoma area of interest (AOI), Table 1 contains the projection information, and Table 2 contains a complete list of analytic deliverables provided to WADNR.

Table 1: Deliverable projection information.

Projections	EPSG	Horizontal Datum	Units
WA State Plane South	NAD83 (HARN)*	NAV88 (GEOID 12b)	US Survey Feet

Table 2: Products delivered to WADNR for City of Tacoma 2024.

Product Type	File Type	Product Details

Vectors	ESRI File Geodatabase (*.gdb) and Shapefiles (*.shp)	• Classified Land Cover
		  ○ Land cover class and description
		• Attributed Individual Tree Segments
		  ○ Tree height, coniferous or deciduous classification, dominant or codominant classification, health
		• Attributed Individual Tree Points
		  ○ Tree height, coniferous or deciduous classification, dominant or codominant classification, health
		• FGDC Compliant Metadata

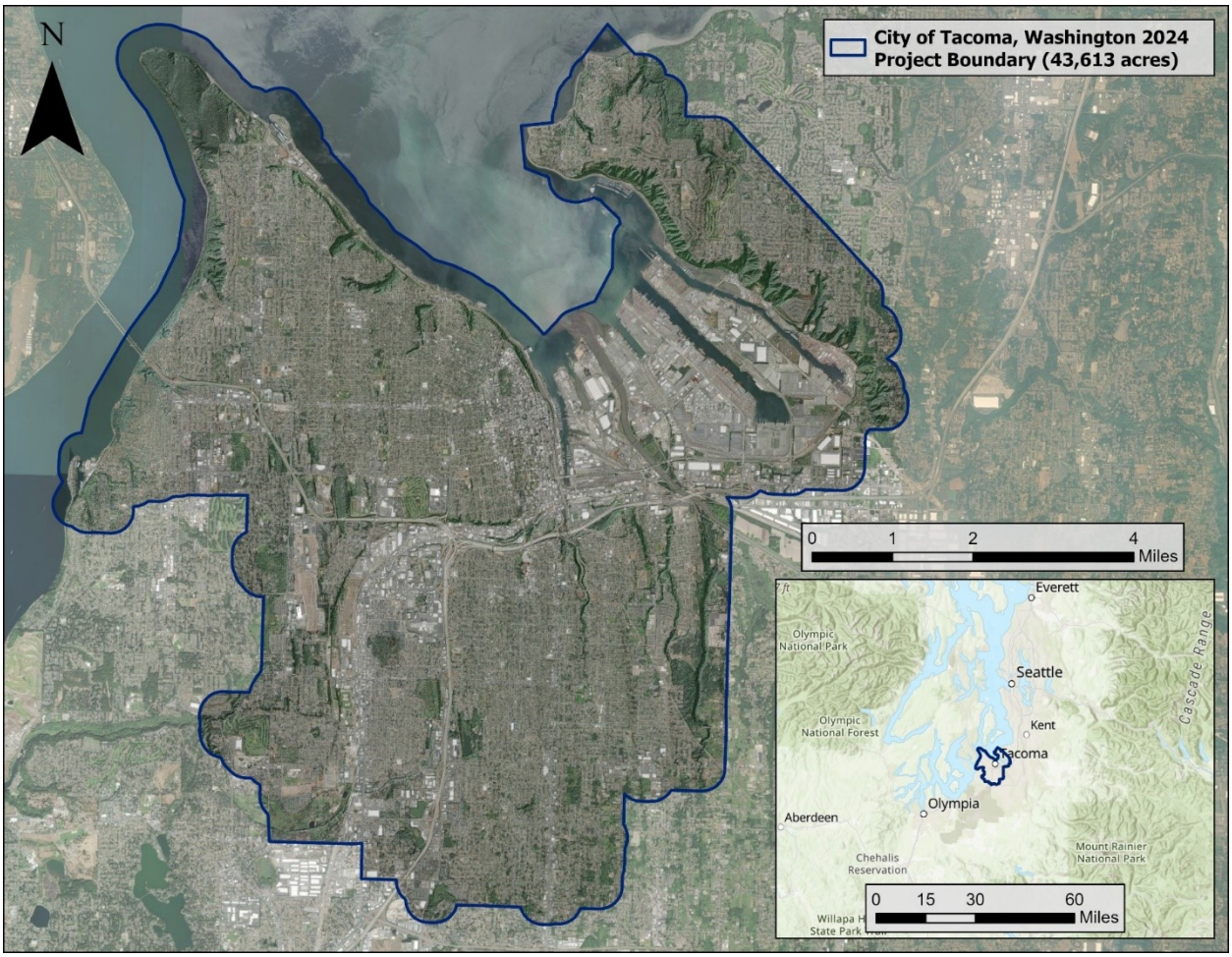

Figure 1: Location map of the City of Tacoma 2024 site in Washington.

2.0 SOURCE DATA

The source data used in the land cover and tree segmentation products includes a colorized and classified lidar point cloud and digital four band imagery (red, green, blue, near infrared).

2.1 LIDAR

The lidar survey was accomplished using a Riegl VQ-1560ii-S system mounted in a Cessna Caravan. Acquisition targeted an average pulse density of at least 8 pulses/m2 over the City of Tacoma project area. The classified point cloud contains ten classes as outlined in Table 3.

Table 3: Lidar classification scheme as determined by the Lidar Base Specifications 2.1 specifications.

Class	Classification Name	Description
1	Default/Unclassified	Laser returns that are not included in the ground class, composed of vegetation and anthropogenic features
1W	Edge Clip / Withheld	Unclassified points flagged as withheld. These are primarily "edge" points from higher scan angles and deemed geometrically unreliable.
2	Ground	Laser returns that are determined to be ground using automated and manual cleaning algorithms
5	High Vegetation	Vegetation greater than 10-ft above ground
6	Buildings	Permanent structures such as buildings that are 10-ft^2 and above
7W	Low Noise / Withheld	Laser returns that are often associated with artificial points below the ground surface
9	Water	Laser returns that are determined to be water using automated and manual cleaning algorithms
17	Bridge	Bridge decks
18W	High Noise / Withheld	Laser returns that are often associated with birds, scattering from reflective surfaces
20	Ignored Ground	Ground points proximate to water's edge breaklines; ignored for correct model creation

2.1.1 Lidar Colorization

The LAS point data was colorized using a process developed by NV5 to reduce occlusion and lean of above ground features, creating true vertical alignment of the imagery with the lidar data. The origin point of each lidar pulse is extracted from the smoothed best estimate of trajectory (SBET) and associated with the most proximate image capture location derived from the exterior orientation of the camera. Each pulse is then assigned spectra from the image with the most similar look angle thereby eliminating seam lines and allowing for selective assignment from overlapping images. An example of the colorized LAS is shown in Figure 2.

Figure 2: Colorized lidar point cloud 3D visualization.

2.2 IMAGERY

Following LAS colorization, 4-band raster images were generated using first return points sampled at a 1.5ft resolution. Imagery developed from the colorized LAS data (ray-traced imagery) is slightly more aesthetically chaotic than traditional orthoimagery but has more precise agreement with above ground features to aid in individual tree analysis.

For land cover and tree segmentation analysis purposes, the 0.25 foot, 4-band orthoimagery was only used for photo interpretation reference. All modeling inputs were generated from the ray-traced imagery dataset.

Figure 3 raytraced imagery (left) and Orthoimagery (right)

3.0 METHODS

3.1 LAND COVER CLASSIFICATION

A seven-class land cover product was created for this project. Land cover classes are detailed in Table 4. The land cover product was generated using object-based image analysis. ~~Incorporating ancillary datasets to aid in the object classification process~~Ancillary datasets were incorporated into the analysis to aid in the object classification process. The ancillary datasets used were a rasterized lidar point classification, lidar height products, imagery derived vegetation indices, publicly available open street maps layers, and publicly available impervious Ecopia 3-ft raster data (2021-2022).

Automated clean-up routines were developed to ensure feature continuity, reduce noise, and classify streets. A quality assurance process, consisting of a team of trained photo interpreters, was used to ensure that all data were appropriately classified at a 1:1000 scale. Where the automated classification failed, heads-up digitization was used to correct the appropriate classification. The minimum mapping unit for the land cover classification is 200 square feet.

The land cover product follows a canopy dominant classification scheme where features are classified from an aerial perspective. Therefore, features that are occluded by canopy will be mapped as the canopy class.

Table 4: Land cover classification scheme and definitions.

Class	Classification Name	Description
1	Canopy	Tree canopy taller than 10 feet.
2	Non-Canopy Vegetation	Green vegetation shorter than 10 feet.
3	Building	Permanent structures such as buildings that are 10-ft^2 and larger.
4	Other Impervious	General impervious class for non-building and non-road features.
5	Water	Natural water bodies, pools, and fountains.
6	Soil and Dry Vegetation	Soil and dry vegetation, generally at or around ground level.
7	Roads	Roads not including sidewalks.

3.2 TREE SEGMENTATION

Tree canopies with a minimum height of 10ft throughout the project area were detected and delineated using a deep learning approach. The tree segmentation model was trained on structural information from the lidar as well as manually digitized tree polygons created by trained photo interpreters. Once the model was applied across the project area, the predicted tree centers were

iteratively 'grown' and allowed to spatially compete using a cellular automata process[1]. This gives them a natural look and complete spatial coverage of the tree canopied portions of the AOI. For dispersed trees, canopy segments correspond to an individual tree crown, however within dense forests they will generally capture the dominant and co-dominant tree crowns. Each detected tree is characterized by a polygon representing the crown extent and a point representing the tree top (or center in the case of complex deciduous canopies).

3.3 TREE ATTRIBUTION

Each canopy identified during segmentation was attributed with additional feature labels including tree height, health, deciduous or coniferous, and dominant or codominant.

3.3.1 Deciduous and Coniferous Classification

In support of the deciduous and coniferous classification, lidar-derived structural metrics were calculated from the classified point cloud for each tree. ~~Photointerpreters~~Photo interpreters created training data for deciduous and coniferous trees, aided using 0.25ft orthoimagery, a canopy height model, and Google Street View. Next, photo interpreted reference trees were matched with tree polygons generated using the segmentation model.

The reference tree information was used to train machine learning models to predict each tree crown as deciduous or coniferous.

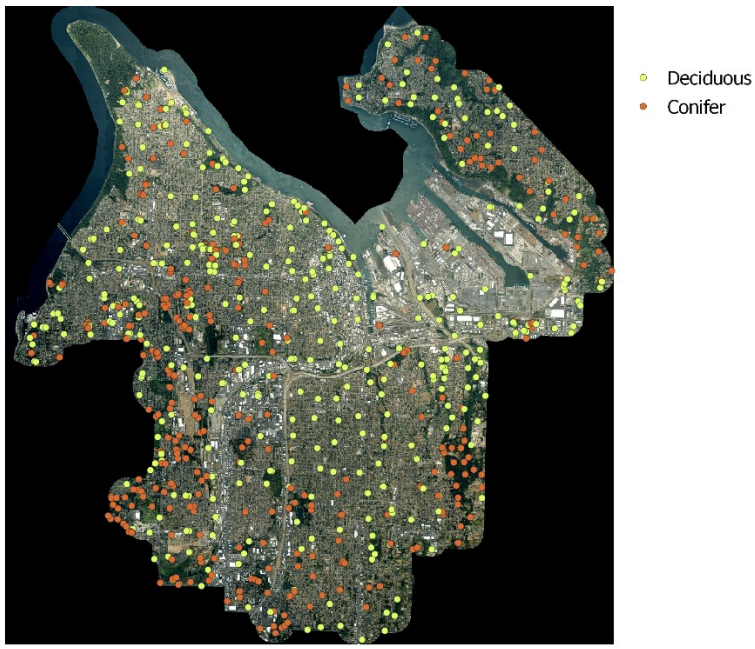

Figure 4 Photo interpreted training points.

[1] Crooks, A.T. 2017. Cellular Automata, in Richardson, D., Castree, N., Goodchild, M. F., Kobayashi, A. L., Liu, W. and Marston, R. (eds.), The International Encyclopedia of Geography: People, the Earth, Environment, and Technology, Wiley Blackwell. DOI: 10.1002/9781118786352.wbieg0578.

3.3.2 Health Assessment

Three categories were used to classify tree health: Low vitality, Normal vitality, and High vitality. Imagery derived vegetation indices (Table 5) were calculated and summarized within each tree polygon then index thresholds were used to establish which health class each canopy belonged to (Table 6).

Table 5: Formula used for the imagery derived vegetation index.

Index	Acronym	Formula
Normalized Difference Vegetation Index	NDVI	(NIR - R) / (NIR + R)

Table 6: NVDVI health ranges.

Low vitality - NDVI range	Normal vitality - NDVI range	High vitality - NDVI range
< 0.3	0.3 – 0.5	> 0.5

3.3.3 Canopy Dominance

Canopy dominance was evaluated for each tree segment. Trees were classified as dominant if they were taller than 6ft from direct neighbors or had no direct neighbors. Both conditions would lead to solar exposure to the sides of the tree. All other trees were classified as codominant. Intermediate and subdominant classifications were not employed because the tree segmentation product represents dominant and codominant trees only.

4.0 RESULTS AND DISCUSSION

4.1 LAND COVER CLASSIFICATION

~~4.1~~ The land cover classification was created as a vector product with full coverage of the City of Tacoma area of interest. Figure 5 contains an example of the land cover classification and **Error! Reference source not found.** outlines the land cover composition for the City of Tacoma.

Figure 5 Land Cover classification results on the right and imagery on the left.

Table 7: Land cover composition for the City of Tacoma.

Class	Area in analysis (acres)	Percent area (%)
Tree Canopy	8,865	20.5
Non-Canopy Vegetation	5,116	11.8
Building	5,591	12.9
Road	4,195	9.7
Other Impervious	8,543	19.7
Water	4,563	10.5
Soil and Dry Vegetation	6,442	14.9
Total	43,315	100.00

4.2 TREE SEGMENTATION AND CLASSIFICATION

The individual tree segmentation routine resulted in approximately 772,144 trees (over 10ft tall) identified across the 43,614 acres that were assessed as a part of this project. Figure 6 shows an example of the individual tree segmentation results overlain on the aerial imagery and symbolized using the canopy height model. Approximately 79% of the trees identified were predicted as deciduous with the remaining 21% predicted as coniferous trees. Additionally, 23% of the trees identified classified as dominant with the remaining 77% classified as codominant trees. Tree heights were assigned in feet based on the 95th percentile of the point cloud data, rather than the maximum height, to minimize the influence of high outliers and noise. Consequently, some shorter trees (around 10ft max height) may be attributed with slightly lower heights due to this percentile-

based approach.

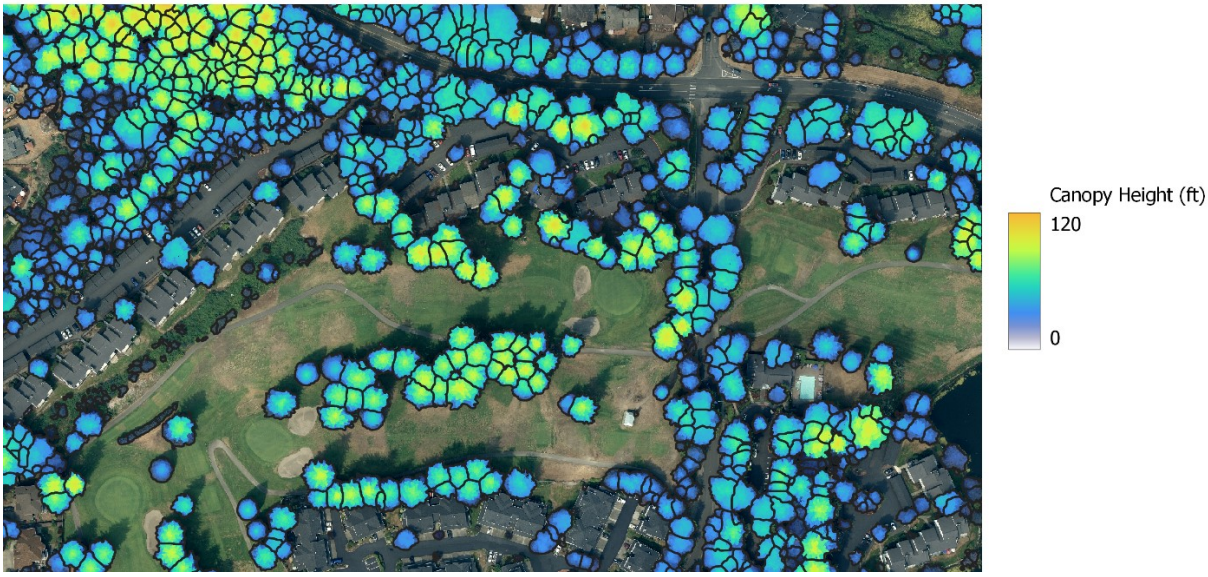

Figure 6: Example region of the individual tree segmentation results.

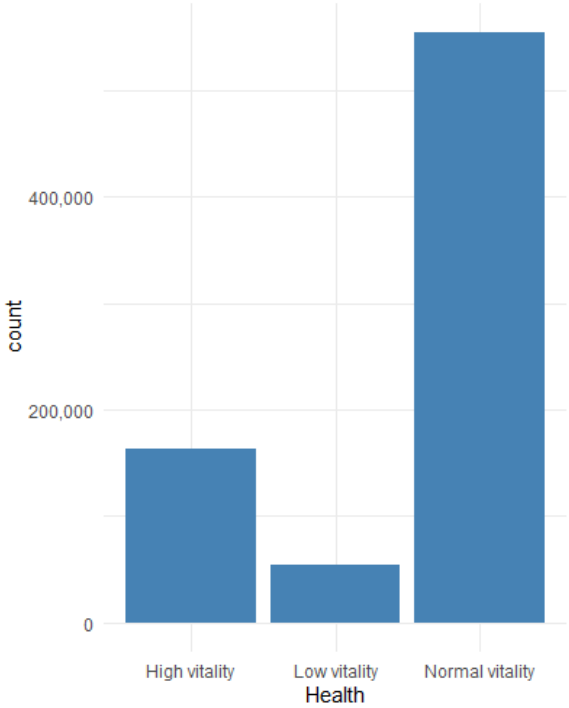

Figure 7: Bar chart showing the results of the health classification of all trees in the analysis.

5.0 APPENDIX C: DATA DICTIONARIES

Table 8: Data dictionary for the land cover product (Land_Cover.gdb).

Field	Unit	Description
OBJECTID	-	ESRI required field that is a unique, not null integer used to identify table rows
Class	-	Class name
Descript	-	Class description
Shape_Length	Feet	ESRI required field populated with the perimeter length of polygon feature
Shape_Area	Sq ft	ESRI required field populated with the area of the polygon feature

Table 9: Data dictionary for the individual tree polygons (tree_crowns.shp).

Field	Unit	Description
FID	-	ESRI required field that is a unique, not null integer used to identify table rows
treeID	-	Unique tree identifier
Height_ft	Feet	95th percentile of lidar height
Conifer	-	Binary value indicating whether a tree is deciduous or coniferous. TRUE= coniferous; FALSE= deciduous.
Health	-	Health classification. Possible values: Low vitality, Normal vitality, High vitality
Dominant	-	Binary value indicating whether a tree is dominant or codominant. TRUE= dominant; FALSE= codominant.

Table 10: Data dictionary for the individual tree points (tree_points.shp).

Field	Unit	Description
FID	-	ESRI required field that is a unique, not null integer used to identify table rows
treeID	-	Unique tree identifier
Height_ft	Feet	95th percentile of lidar height
Conifer	-	Binary value indicating whether a tree is deciduous or coniferous. TRUE= coniferous; FALSE= deciduous.
Health	-	Health classification. Possible values: Low vitality, Normal vitality, High vitality
Dominant	-	Binary value indicating whether a tree is dominant or codominant. TRUE= dominant; FALSE= codominant.